

Magnetometer Sensor

Objective:

To read and analyze the magnetic field values along the X, Y, and Z axes using the built-in LSM9DS1 magnetometer sensor on the Arduino Nano 33 BLE.

Procedure:

1. Opened Arduino IDE.
2. Installed the Arduino_LSM9DS1 library.
3. Opened the SimpleMagnetometer example from:
4. File → Examples → Arduino_LSM9DS1 → SimpleMagnetometer
5. Uploaded the program to the Arduino Nano 33 BLE board.
6. Opened the Serial Monitor at 9600 baud rate.
7. Moved the board near metallic objects or changed its orientation.
8. Observed the magnetic field values displayed in the Serial Monitor.

Working Principle:

The magnetometer measures the magnetic field around the board along three axes:

- X-axis → Magnetic field in X direction
- Y-axis → Magnetic field in Y direction
- Z-axis → Magnetic field in Z direction

It works like a digital compass by sensing Earth's magnetic field.

When the board is rotated, the magnetic field values change.

Observations:

- Values changed when the board orientation was changed.
- Magnetic field readings varied near metal objects and electronic devices.

Different axes produced different values depending on the direction of the magnetic field.

-9.22	30.46	-0.85
-3.78	41.50	13.92
-0.61	50.54	24.96
1.34	54.08	32.23
4.82	54.57	36.13
9.52	56.27	38.39
14.16	56.40	37.54
17.46	49.99	33.75